# 3SP-DUNet: An Interpretable Deep-Unfolded Network With Spatial–Spectral Sparse Priors for Hyperspectral Pansharpening

Yong Yang, *Senior Member, IEEE*, Xuan Liu, Shuying Huang, *Member, IEEE*, Xiaozheng Wang, and Jinlin Lu

***Abstract*—Most existing deep learning-based pansharpening methods adopt typical black-box models to achieve the fusion of panchromatic (PAN) and low-resolution hyperspectral (LRHS) images. This type of architecture has limited interpretability, making it difficult to ensure the fidelity of spatial and spectral features in the fusion results. To address this issue, we propose an interpretable deep unfolding network with spectral–spatial sparsity priors (3SP-DUNet), which is designed based on the unfolding optimization algorithm for solving the HS pansharpening problem. First, the objective function for solving high-resolution hyperspectral (HRHS) images in the HS pansharpening task is optimized to obtain the optimized solution formula. Then, based on the solution terms in the formula, spectral degradation/reconstruction (SpeD/SpeR) blocks and spatial degradation/reconstruction (SpaD/SpaR) blocks are constructed to simulate the degradation and reconstruction process of spectral and spatial features. Regarding the prior regularization terms in the formula, spectral/spatial sparse prior blocks are designed to constrain the solution space, ensuring the fidelity of spectral and spatial features. In addition, to reduce feature loss caused by increasing network depth, a cross-stage memory fusion unit (CMFU) is constructed to supplement features by fusing reconstructed images from adjacent stages. To guide the model to reconstruct accurate spectral features, a spectral similarity loss is defined by calculating the cosine values of the corresponding spectral vectors between the fused image and the corresponding ground-truth (GT) image. Extensive experiments demonstrate that 3SP-DUNet surpasses some state-of-the-art (SOTA) methods in both quantitative metrics and visual quality, while offering superior interpretability. The source code of this work will be released on GitHub.**

***Index Terms*—Deep unfolding, hyperspectral (HS) pansharpening, interpretable deep learning (DL), sparse representation.**

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Yong Yang is with the School of Computer Science and Technology, Tiangong University, Tianjin 300387, China, and also with Cangzhou Institute of Tiangong University, Cangzhou 061000, China (e-mail: greatyangy@126.com).

Xuan Liu, Xiaozheng Wang, and Jinlin Lu are with the School of Control Science and Engineering, Tiangong University, Tianjin 300387, China (e-mail: liuxuan2brilliant@outlook.com; xiaozhengwang95@gmail.com; jinlinlu03@163.com).

Shuying Huang is with the School of Software, Tiangong University, Tianjin 300387, China, and also with Cangzhou Institute of Tiangong University, Cangzhou 061000, China (e-mail: shuyinghuang2010@126.com).

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## I. Introduction

CAPTURING high-resolution hyperspectral (HRHS) images with a single sensor is a significant challenge in optical remote sensing. Hyperspectral (HS) sensors can capture rich spectral information in hundreds of narrowbands, generating HS images with high spectral resolution and low spatial resolution. Conversely, panchromatic (PAN) sensors can collect information across the entire wavelength range, thereby obtaining PAN images with high spatial resolution and low spectral resolution. By fusing PAN images with low-resolution HS (LRHS) images, HS pansharpening techniques can generate HRHS images with high spatial and spectral fidelity to improve the performance of subsequent high-level computer vision tasks [1], [2], [3], such as environmental monitoring, target detection, and classification [4], [5], [6], [7].

Traditional HS pansharpening methods are generally categorized into three classes [8]: component substitution (CS)-based [9], [10], [11] multiresolution analysis (MRA)-based [12], [13], [14], and variational optimization (VO)-based methods [15], [16], [17]. CS-based methods aim to extract specific components from LRHS and PAN images using reversible projection algorithms and reconstruct the HRHS image by substituting or fusing these components. MRA-based methods employ multiresolution techniques to extract high-frequency spatial details from the PAN image, and then inject them into spectrally adjusted HS components to enhance the spatial resolution of LRHS images. Compared to CS methods, MRA approaches typically yield better spectral fidelity. However, designing suitable spatial filters across multiple scales is a challenging task that makes the implementation of MRA methods relatively more complex. VO-based methods reconstruct HRHS images by solving an optimization problem that requires manually defining regularization terms using domain prior knowledge to improve the accuracy of the solution. Although this type of method is theoretically interpretable, manually designing prior terms often requires specialized domain knowledge and lacks flexibility, which also limits its performance in practical applications [18], [19].

Recently, deep learning (DL)-based approaches have become dominant in the HS pansharpening task. Unlike traditional methods that rely on handcrafted priors [20], [21], DL-based models, such as convolutional neural networks (CNNs) [13], [22], [23], generative adversarial networks

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(GANs) [24], [25], and graph neural networks (GNNs) [26], directly learn the complex nonlinear mapping from LRHS and PAN images to the desired HRHS image. For instance, HyperDSNet [27] achieved the fusion of deep and shallow features by combining multidetail extractors with spectral attention, while HyperRefiner [28] introduced adaptive cross attention to improve the accuracy of feature extraction. FMPM-DNet [29] used dynamically modulated units to fuse features and employed a probabilistic mask to embed fine-grained details. Other works have explored more complex hierarchical structures. Tree-SNet [30] constructed a spectral–spatial tree with adaptive convolutions, while recent Transformer-based models like DCTransformer [31] and DFCFN [32] leveraged advanced cross-attention mechanisms and feature enhancement backbones for more robust fusion. AF2GNN [33] introduced a GNN based on adaptive filters and multiaggregator fusion, which significantly enhances feature representation in heterogeneous regions by adaptively combining distinct filters. Similarly, DHMG [34] presented a collaborative multi-GNN framework utilizing hybrid filters to simultaneously capture spectral fidelity and spatial details. Furthermore, SLCGC [35] accurately modeled the intrinsic structure of HS data by dynamically adjusting both graph topology and frequency response. Despite these advancements yielding impressive performance, most of these models are still black-box models that lack physical interpretability and are prone to spatial and spectral distortions in fusion results.

In recent years, to provide theoretical interpretability and robustness for networks, some researchers have conducted research on deep unfolding networks (DUNs) to bridge the gap between model-based iterative optimization and data-driven DL. For instance, Yan et al. [36] incorporated a low-rank prior into an optimization model and constructed a DUN based on the alternating direction method of multipliers (ADMM). Similarly, Zhang et al. [37] unfolded the iterations of the proximal gradient descent algorithm into a DUN, integrating a GNN to exploit spectral–spatial priors. Furthermore, Ma et al. [38] developed a DUN grounded in proximal gradient descent, incorporating a Transformer architecture to capture implicit priors. Although these methods integrate the prior knowledge of traditional methods into deep networks, they do not fully consider the complex degradation factors that exist in two modal images, resulting in inaccurate reconstruction of spectral and spatial features.

Based on the above issues, combining the interpretability of traditional methods with the powerful feature extraction capability of DL-based methods, we propose a new interpretable DUN with spatial–spectral sparse priors (3SP-DUNet) for the HS pansharpening task. Specifically, for the observation model in traditional HS pansharpening methods, we decouple the HS pansharpening solving problem into two subproblems using the half-quadratic splitting (HQS) method. Then, based on the optimization process of the two subproblems, 3SP-DUNet is constructed to reconstruct the HRHS image. In 3SP-DUNet, corresponding learnable modules are constructed to solve the spectral and spatial fidelity terms, as well as the prior regularization terms in the optimization formula for two subproblems. Our model not only has good interpretability but also can reconstruct HRHS images with high spatial and spectral fidelity. The main contributions of this work are summarized as follows.

1) We propose a 3SP-DUNet based on the unfolding optimization process for the HS pansharpening problem to achieve the fusion of PAN and LRHS images. Each module within the network is built based on specific physical principles, thereby ensuring high transparency and interpretability of the model.
2) Spatial degradation/reconstruction (SpaD/SpaR) and spectral degradation/reconstruction (SpeD/SpeR) modules are constructed to simulate the respective degradation and reconstruction processes of spatial and spectral features, in order to solve the spatial and spectral fidelity terms in the unfolding optimization algorithm.
3) Spatial/spectral sparse prior blocks based on sparse representation learning are designed to address the sparse prior regularization terms by leveraging latent spatial and spectral information to reconstruct the PAN and HRHS images at each stage of the model.
4) A cross-stage memory fusion unit (CMFU) is constructed to selectively integrate the outputs of adjacent stages in the model, thereby mitigating the loss of spatial and spectral information during the iterative process.

## II. Related Works

### A. Traditional HS Pansharpening Methods

The CS-based [9], [10], [11], MRA-based [12], [13], [14], and VO-based [15], [16], [17] approaches are all designed according to the observation model and address the HS pansharpening problem by constructing optimized solution frameworks. The observation model [39], [40], [41] can be formulated as follows:

$$\begin{aligned} Z &= XB + N_H \\ Y &= SX + N_P \end{aligned} \tag{1}$$

where $X \in \mathbb{R}^{C\times H\times W}$ denotes a target HRHS image with C spectral bands and a spatial size of $H \times W$, $Z \in \mathbb{R}^{C\times h\times w}$, and $Y \in \mathbb{R}^{1\times H\times W}$ represent the observed LRHS image and PAN image, respectively. $B \in \mathbb{R}^{HW\times hw}$ denotes the spatial response function of the HS sensor, and $S \in \mathbb{R}^{C\times 1}$ represents the spectral response of the PAN sensor. $N_H$ and $N_P$ are the additive noise in the LRHS and PAN images, respectively.

To improve the performance of pansharpening tasks, many approaches constrain the solution space for the target image $X$ by constructing regularization terms based on image prior information. Palsson et al. [42] proposed a pansharpening method based on total variation (TV), which uses TV to regularize the pansharpening problem dictated by an explicit image formation model. An orthogonal TV analysis (OTVCA) method [43] was proposed to achieve feature extraction of remote sensing HS data by minimizing a TV penalty optimization problem. Although TV regularization terms can effectively preserve the edges of the image, they can also smooth out some small texture information. To reduce the loss of spatial and spectral information, collaborative nonnegative matrix factorization (CNMF) [44] was introduced for LRHS and MS

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image fusion. However, these traditional methods that rely on manually defined priors often involve challenging parameter estimation processes, and inaccurate parameter estimation may lead to spectral or spatial distortions, thereby limiting their applicability in practical scenarios.

### B. DUN for Pansharpening

DUN, which combines traditional iterative optimization algorithms with DL techniques, aims to learn the mapping relationship between network inputs and outputs by constructing theoretically interpretable networks. This type of network can avoid the artificial definition of priors in traditional methods and fully utilizes the network's learning ability, therefore, it has a better performance and generalization ability compared to general black-box networks and traditional methods. DUNs have achieved promising results in diverse inverse problems, such as image super-resolution reconstruction and compressed sensing tasks.

In the HS pansharpening task, a type of network is constructed by focusing on introducing powerful image priors to guide the fusion process. HyperPNN [45] designed a spectral prediction CNN to learn and represent the spatial–spectral prior, thereby improving the spectral prediction ability of the model. DIP-HyperKite [46] learned deep spectral–spatial priors jointly through stacked architectures, and proposed a learnable spectral response function (SRF) to estimate the PAN image from the upsampled HS image. GuideNet [47] was designed as a universal high-resolution guided CNN framework that uses spatial details as a prior term to drive the injection of high-resolution MS information into the feature domain. DHP-DARN [48] utilized deep HS prior (DHP) to enhance the model's ability to identify different features. VaFuNet [49] constructed a DUN based on the VO model, employing a data-driven approach to capture prior information. Similarly, SMGU-Net [50] introduced a spectral–spatial mutually guided unfolding framework that learns priors via data-driven methods, achieving competitive performance. The success of these approaches demonstrates that the effective integration of spectral and spatial priors is a critical factor for achieving state-of-the-art (SOTA) performance in DUN-based HS pansharpening methods.

Another category of DUNs introduces the idea of CS or MRA to optimize spectral and spatial feature representations. For instance, MD$^3$Net [51] incorporated a deep residual connection module to exploit spectral–spatial correlations within multiscale data. DIM-FuNet [52] integrated a fusion mechanism to extract and learn detail injection components. In addition, IU2PNet [53] employed a deep feature pyramid to enhance cross-resolution information interaction. However, methods have a problem in that they typically directly fuse the features of two different modal images without considering the modal differences between the two image features, which inevitably introduces spectral and spatial distortions.

## III. Proposed Method

In this section, we propose a DUN based on the unfolding optimization solution process for the observation model, called 3SP-DUNet, as shown in Fig. 1. First, the solution process for the HS pansharpening model is optimized. Then, by analyzing the derived solution formulas, we design a 3SP-DUNet consisting of multiple fusion stages to iteratively solve the HRHS image. Between two adjacent stages, a CMFU is designed to reduce the loss of spectral and spatial information. Below, we present the optimization process of the HS pansharpening model and the structure of 3SP-DUNet.

### A. Model Optimization

Estimating HRHS images from observed LRHS and PAN images is an ill-posed inverse problem. According to the observation model [see (1)], the objective function for reconstructing HRHS images in HS pansharpening tasks is usually defined as follows:

$$\underset{X}{arg\,\min}\ \frac{1}{2}\|Z - XB\|_F^2 + \frac{1}{2}\|Y - SX\|_F^2 + \lambda J(X) \tag{2}$$

where the first and second terms represent the spectral and spatial fidelity terms, respectively, and the third term represents the regularization term. The parameter $\lambda$ is a weighting factor to balance the fidelity terms and the regularization term, and $\|\cdot\|_F$ represents the Frobenius norm.

To solve (2), we employ the HQS method. By introducing an auxiliary variable $V$ ($V \approx X$), (2) can be reformulated as an unconstrained optimization problem

$$\underset{X,V}{arg\,\min}\left(\frac{1}{2}\|Z - XB\|_F^2 + \frac{1}{2}\|Y - SX\|_F^2 + \xi\|V - X\|_F^2 + \lambda J(V)\right) \tag{3}$$

where $\xi$ is a penalty parameter that controls the degree of consistency between $X$ and $V$. To facilitate a tractable solution, the optimization problem in (3) is reformulated via a variable decoupling strategy involving $X$ and $V$. This decomposition yields two distinct subproblems solved via alternately minimizing the two variables. Specifically, by treating $V$ as a fixed parameter, the optimization is simplified to a minimization problem with respect to $X$, expressed as follows:

$$X^{(t+1)} = \underset{X}{arg\,\min}\ \frac{1}{2}\|Z - XB\|_F^2 + \frac{1}{2}\|Y - SX\|_F^2 + \xi\left\|V^{(t)} - X\right\|_F^2. \tag{4}$$

Analogously, following the same derivation logic, the terms associated with $V$ are isolated while holding $X$ constant, yielding the corresponding subproblem

$$V^{(t+1)} = \underset{V}{arg\,\min}\,\xi\left\|V - X^{(t+1)}\right\|_F^2 + \lambda J(V) \tag{5}$$

where $t$ represents the number of iterations.

Subsequently, a gradient descent strategy is employed to solve the subproblems defined in (4) and (5). Consequently, the iterative update rules for the target variable $X$ and the auxiliary variable $V$ are formulated as follows. Specifically,

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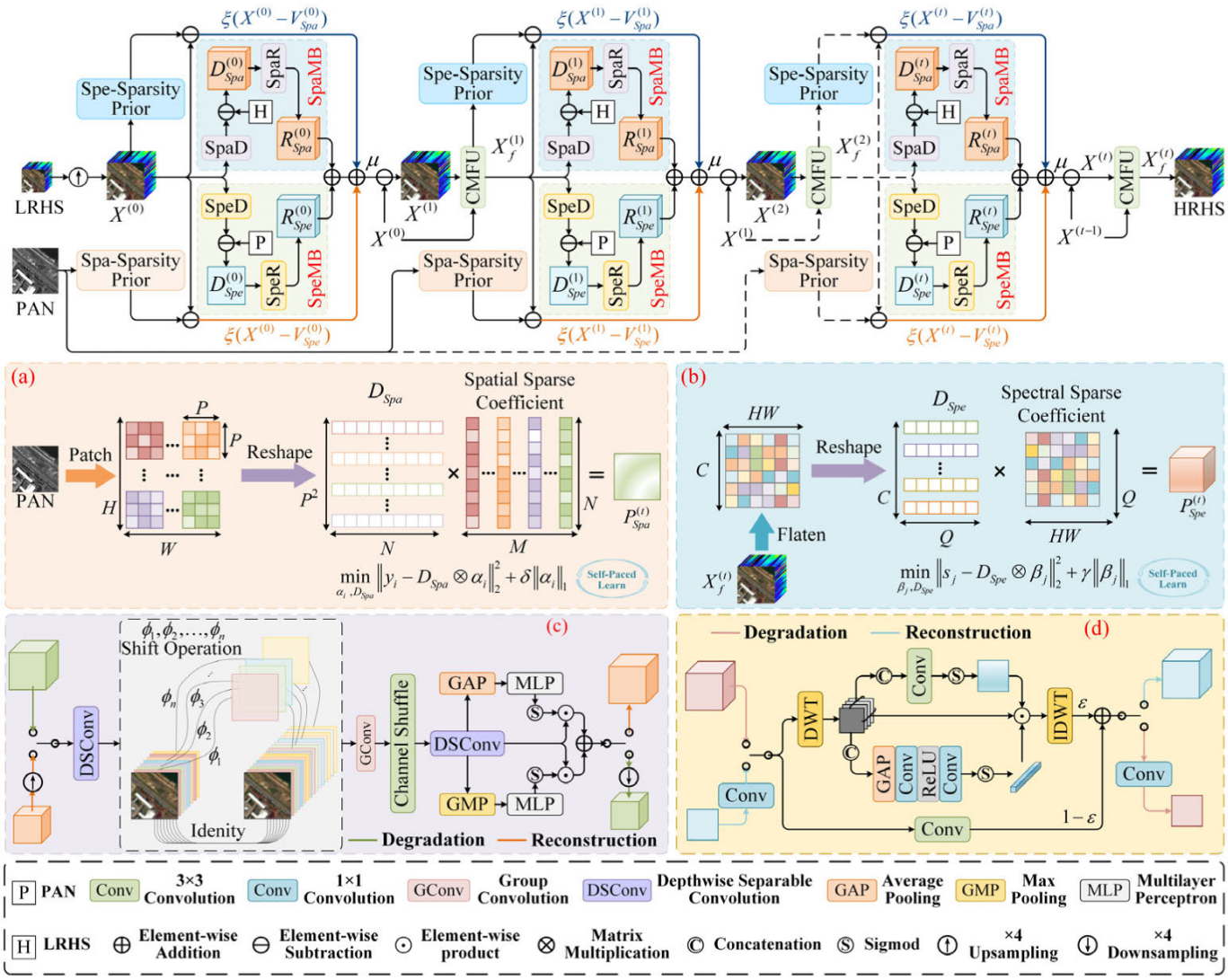



Fig. 1. Overall architecture of 3SP-DUNet. (a) Spa-sparsity prior. (b) Spe-sparsity prior. (c) SpaD/SpaR. (d) SpeD/SpeR.

(6) is derived by applying a gradient descent step with step size $\mu$ to the objective function in (4)

$$X^{(t+1)} = X^{(t)} - \mu \left[ \begin{array}{l} \left(X^{(t)}B - Z\right)B^T \\ \quad + S^T\left(SX^{(t)} - Y\right) + 2\xi\left(X^{(t)} - V^{(t)}\right) \end{array} \right] \tag{6}$$

$$V^{(t)} = prox_{J(\cdot)}\left(X^{(t-1)}\right) \tag{7}$$

where $\mu$ is the iteration step size, which is a learnable parameter. $prox_{J(\cdot)}$ denotes the proximal operator associated with the implicit prior $\lambda J(\cdot)$, and $T$ indicates the transpose of a matrix. At each iteration, $V$ is updated first, followed by $X$.

To solve (6) and (7), traditional methods typically require manually setting parameters based on experience, such as $B, S, \mu$, and $\xi$, which directly affect the accuracy of the solution. In this article, we construct a 3SP-DUNet based on (6) and (7) to solve the HRHS images, $X$.

### B. 3SP-DUNet

Equations (6) and (7) aim to infer an HRHS image by alternately optimizing the subproblems about $X$ and $V$. Based on the two formulas, we construct a 3SP-DUNet consisting of multiple iterative solution stages, as shown in Fig. 1. Equation (6) can be divided into four subterms, each corresponding to a module in each stage of the network. Specifically, $S^T(SX^{(t)}-Y)$ is implemented by SpeMB, while $(X^{(t)}B-Z)B^T$ is implemented by SpaMB. These two blocks can refine spectral and spatial features by simulating sensor-specific spectral and spatial perceptual degradation mechanisms. $V$ is an auxiliary variable that approximates the variable $X$, which means that $V$ needs to have both rich spatial and spectral information. Therefore, we split the constraint term $2(X^{(t)}-V^{(t)})$ into two terms $(X^{(t)}-V_{Spa}^{(t)})$ and $(X^{(t)}-V_{Spe}^{(t)})$ to constrain spatial and spectral information, respectively. Corresponding to $V_{Spa}^{(t)}$ and $V_{Spe}^{(t)}$ in these two constraint terms, (6) needs to be solved for the PAN image $Z$ and the solution $X^{(t)}$ at each iteration. The solution of $V$ in (7) corresponds to two modules in the network, namely the spatial and spectral sparse priors. Below, we provide a detailed introduction to the structure of the $t$th stage in the network.

First, spatial and spectral sparse priors are applied to obtain $V_{Spa}^{(t)}$ and $V_{Spe}^{(t)}$. To constrain the reconstructed results to

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preserve more spatial and spectral information, the input PAN image $Z$ is directly used to construct the spatial sparsity prior. To ensure that the reconstructed results retain richer spectral information, a CMFU is executed to achieve the fusion of the reconstruction results $X^{(t)}$ and $X^{(t-1)}$ from the previous two stages. The fusion result $X_f^{(t)}$ is then used to construct a spectral sparse prior. The above operations are as follows:

$$V_{Spa}^{(t)} = SpaSP(Z), \quad V_{Spe}^{(t)} = SpeSP\left(X_f^{(t)}\right) \tag{8}$$

$$X_f^{(t)} = CMFU\left(X^{(t)}, X^{(t-1)}\right) \tag{9}$$

where $SpaSP(\cdot)$ and $SpeSP(\cdot)$ denote the operations of spatial and spectral sparse priors, respectively, and $CMFU(\cdot)$ denotes the operation of CMFU.

Then, the fused image $X_f^{(t)}$ is fed into SpeMB and SpaMB to solve $S^T(SX_f^{(t)} - Y)$ and $(X_f^{(t)}B - Z)B^T$. Note that the input $X_f^{(0)}$ in the first stage is the upsampled LRHS image. In SpeMB, a SpeD is designed to learn the spectral degradation operation $SX_f^{(t)}$. SpeR, which has a similar structure to SpeD, represents the spectral reconstruction process, i.e., $S^T(SX_f^{(t)} - Y)$. In SpaMB, SpaD is designed to learn the spatial degradation process operation $X_f^{(t)}B$, while the spatial reconstruction process $(X_f^{(t)}B - Z)B^T$ is implemented using SpaR with a structure similar to SpaD. The operations of SpeMB and SpaMB can be represented by the following equations:

$$R_{Spe}^{(t)} = SpeR\left(SpeD\left(X_f^{(t)}\right) - Y\right) \tag{10}$$

$$R_{Spa}^{(t)} = SpaR\left(SpaD\left(X_f^{(t)}\right) - Z\right) \tag{11}$$

where $SpeD(\cdot)$ denotes the operation of SpeD. $SpeR(\cdot)$ denotes the operation of S{\!}peR, and its output is $R_{Spe}^{(t)}$. $SpaD(\cdot)$ denotes the operation of SpaD. $SpaR(\cdot)$ denotes the operation of SpaR, and its output is. Meanwhile, two constraint terms $(X_f^{(t)} - V_{Spa}^{(t)})$ and $(X_f^{(t)} - V_{Spe}^{(t)})$ are computed, and the hyperparameter $\xi$ is learned.

Finally, according to (6) and (8)–(11), the reconstruction result of the $t$th stage is obtained, which is defined as follows:

$$X^{(t+1)} = X^{(t)} - \mu\left(\begin{array}{c} R_{Spe}^{(t)} + R_{Spa}^{(t)} + \xi\left(X_f^{(t)} - V_{Spa}^{(t)}\right) \\ + \xi\left(X_f^{(t)} - V_{Spe}^{(t)}\right) \end{array}\right). \tag{12}$$

Below, we provide a detailed introduction to the modules adopted in each stage, such as spatial and spectral sparse priors, CMFU, SpeD/SpeR, and SpaD/SpaR.

*1) Spatial/Spectral Sparse Priors:* Compared to traditional methods, sparse representation learning provides a more flexible and data-driven prior learning approach that can reveal the underlying structure and patterns of data by extracting sparse feature representations from the data. Therefore, we construct two modules, spatial and spectral sparse priors, to solve the auxiliary variable $V$ by mining potential spectral and spatial features from the HRMS image obtained at each stage and the input PAN image. The solved $V$ is used to calculate the regularization terms in (6). The specific process of spatial and spectral sparse priors is as follows.

*a) Spatial sparse prior:* As shown in Fig. 1(a), spatial sparse prior aims to learn an overcomplete dictionary $D_{Spa}(\in R^{p^2\times N})$ in which each atom represents a localized spatial structure. After sparse encoding, these atoms can capture the spatial prior required for reconstructing a high-quality HRHS image. Specifically, the PAN image is first divided into a set of nonoverlapping image blocks, each of which is flattened into a vector $y_i \in \mathbb{R}^{p^2\times 1}$, where $p$ denotes the patch size. For each patch $y_i$, a corresponding sparse code $\alpha_i$ is learned. Thus, each block $y_i$ can be approximately reconstructed by $D_{Spa}\otimes\alpha_i$. $\otimes$ represents the matrix multiplication. The optimization problem of spatial sparse prior can be represented by the following formula:

$$\min_{\alpha_i, D_{Spa}} \left\|y_i - D_{Spa}\otimes\alpha_i\right\|_2^2 + \delta\left\|\alpha_i\right\|_1 \tag{13}$$

where $\delta$ is a regularization coefficient that controls the sparsity level of the solution, and $\|\cdot\|_2$ and $\|\cdot\|_1$ represent the 2-norm and 1-norm, respectively. To solve this problem, we adopt the iterative shrinkage–thresholding algorithm (ISTA). The sparse code $\alpha$ is updated iteratively as follows:

$$\begin{aligned} \alpha_i^{(t+1)} &= S_{Spa}\left(\alpha_i^{(t)} - \eta_{Spa}\cdot\nabla f\left(\alpha_i^{(t)}\right)\right) \\ \nabla f\left(\alpha_i^{(t)}\right) &= D_{Spa}^T\left(D_{Spa}\alpha_i^{(t)} - y_i\right) \end{aligned} \tag{14}$$

where $\eta_{Spa}$ is the step size, $S_{Spa}(\cdot)$ denotes the soft-thresholding operator, i.e., $S_{Spa}(\cdot) = sign(\cdot)\cdot\max(|\cdot| - \theta_{Spa}, 0)$, and $\theta_{Spa}$ is the thresholding parameter.

Next, the sparse encoding and dictionary are used to reconstruct each image block, i.e., $\hat{y}_i = D_{Spa}\otimes\alpha_i$, which are placed at the spatial position of the original image block $y_i$. The reconstructed image is the solution of $V_{spa}^{(t)}$ in (12).

*b) Spectral sparse prior:* Similar to the calculation of the spatial sparse prior, as shown in Fig. 1(b), the spectral sparse prior is to capture the spatial and spectral features from HS data. Specifically, the values at each spatial position in the fused image $X_f^{(t)}$ are treated as a spectral vector $s_j(\in R^{Ch\times 1})$, where $Ch$ is the number of channels and $j = 1, \ldots, HW$. For each spectral vector $s_j$, a corresponding sparse code $\beta_j$ and an overcomplete dictionary $D_{Spe}(\in R^{Ch\times Q})$ are learned to reconstruct the HRHS features. The optimization problem of spectral sparse prior can be represented by the following formula:

$$\min_{\beta_j, D_{Spe}} \left\|s_j - D_{Spe}\otimes\beta_j\right\|_2^2 + \gamma\left\|\beta_j\right\|_1 \tag{15}$$

where $\gamma$ is a regularization coefficient that controls the sparsity level. To solve this problem, we adopt the ISTA, where the sparse code $\beta_j$ is iteratively updated as follows:

$$\begin{cases} \beta_j^{(t+1)} = S_{Spe}\left(\beta_j^{(t)} - \eta_{Spe}\cdot\nabla f\left(\beta_j^{(t)}\right)\right) \\ \nabla f\left(\beta_j^{(t)}\right) = D_{Spe}^T\left(D_{Spe}\beta_j^{(t)} - s_j\right) \end{cases} \tag{16}$$

where $S_{Spe}(\cdot) = sign(\cdot)\cdot\max(|\cdot| - \theta_{Spe}, 0)$ denotes the soft-thresholding operator with the threshold $\theta_{Spe}$, and $\eta_{Spe}$ is the step size. After convergence, the learned sparse code $\beta_j$ is used to reconstruct the HRHS features using $\hat{s}_j = D_{Spe}\otimes\hat{\beta}_j$, which are placed at the spatial position of the spectral vector $s_j$. The reconstructed image is the solution of $V_{Spe}^{(t)}$ in (12).

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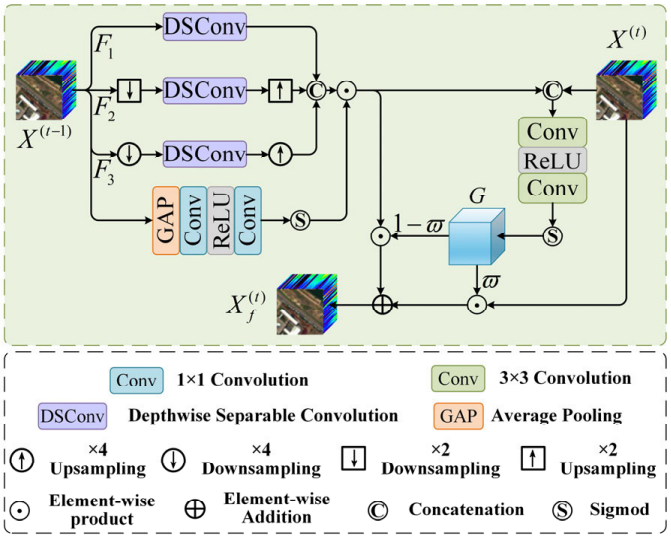



Fig. 2. Architecture of CMFU.

*2) Cross-Stage Memory Fusion Unit:* 3SP-DUNet is a multistage process that gradually refines the fused HRHS images. As the depth of the network increases, there will be a loss of spectral and spatial information. To address these challenges, we build a CMFU, as shown in Fig. 2, which enhances feature propagation across stages by leveraging multiscale feature reorganization and adaptive attention-based gating mechanisms. CMFU consists of two core operations: multiscale feature extraction and dynamic attention-guided modulation.

First, a multiscale feature extraction operation is implemented by performing depthwise separable convolutions on images of different scales to extract multiscale features $F_{i(i=1,2,3)}$ of the reconstructed image from the previous stage. This operation can capture the local texture and global contextual features of the image, which can be represented by the following equation:

$$\begin{aligned} F_1 &= DSConv\left(X^{(t-1)}\right) \\ F_2/F_3 &= \left(DSConv\left(\left(X^{(t-1)}\right)\downarrow_s\right)\right)\uparrow_s \end{aligned} \tag{17}$$

where $DSConv(\cdot)$ represents the depthwise separable convolution with a kernel size of $3 \times 3$, and the operators $\uparrow_s$ and $\downarrow_s$ denote the downsampling and upsampling operations with a scale factor of $s$ ($s = 2$ or 4), respectively.

Then, to better integrate contextual features at different scales, a channel attention mechanism is used to obtain an attention weight matrix, which is used to weight features at three scales to obtain fused features $X_m^{(t-1)}$. The weighted features of the three scales are fused through an addition operation

$$W = S\left(Conv_{1\times1}\left(ReLU\left(Conv_{1\times1}\left(GAP\left(X^{(t-1)}\right)\right)\right)\right)\right) \tag{18}$$

$$X_m^{(t-1)} = \sum_{i=1}^{3} W \odot F_i \tag{19}$$

where $GAP(\cdot)$ represents the global average pooling, and $Conv_{1\times1}(\cdot)$ represents the $1 \times 1$ convolution. $ReLU(\cdot)$ and $S(\cdot)$ represent the ReLU and Sigmoid activation functions, respectively. $\odot$ represents the elementwise multiplication.

Subsequently, to avoid spatial and spectral distortions caused by direct aliasing of image features from two stages, a dynamic attention guided modulation mechanism is designed to selectively fuse the two features by adopting an adaptive gating operation. Specifically, the multiscale fused features $X_m^{(t-1)}$ are connected to the current reconstructed image $X^{(t)}$ and then fed into a series of operations, including a $3 \times 3$ convolution, a ReLU function, a $3 \times 3$ convolution, and a Sigmoid function, to obtain a feature modulation matrix $G$. The matrix $G$ is used to modulate multiscale fused features and the reconstructed image $X^{(t)}$, and these two modulated features are fused together through adaptive feature selection using a learnable parameter $\varpi$. The above operation is as follows:

$$G = S\left(Conv_{3\times3}\left(ReLU\left(Conv_{3\times3}\left(Cat\left(X^{(t)}, X_m^{(t-1)}\right)\right)\right)\right)\right) \tag{20}$$

$$X_f^{(t)} = \varpi\left(X^{(t)} \odot G\right) + (1-\varpi)\left(X_m^{(t-1)} \odot G\right) \tag{21}$$

where $Conv_{3\times3}(\cdot)$ represents the $3 \times 3$ convolution, and $Cat(\cdot)$ represents the concatenation operation.

*3) SpaD/SpaR Blocks:* In the process of solving HS pan-sharpening problems, spatial degradation and reconstruction operations are its core physical processes. The spatial degradation operation simulates the degradation process of spatial information blurring and loss caused by sensor limitations, platform motion, and other factors. On the contrary, the image reconstruction operation maps the features from the low-resolution space back to the high-resolution space. To effectively simulate the process of spatial degradation and reconstruction, we construct adaptive learnable SpaD and SpaR shown in Fig. 1(c) by introducing physically driven shift operations and a channel interaction mechanism to better reflect the spatial changes inherent in the image. The operations of SpaD and SpaR are the same, except for their different inputs and outputs. The input of SpaD is an HS image in high-resolution space, and the output is an HS image in low-resolution space, while the input and output of SpaR are opposite to those of SpaD. Next, we take SpaD as an example to introduce its specific structure.

First, the fused image $X_f^{(t)}$ from CMFU is used as input for SpaD, and depthwise separable convolutions are performed to independently extract local spatial features within each channel. Then, a shift operation is applied to explicitly and randomly displace each feature map in multiple directions (e.g., up, down, left, and right) with different amplitudes to simulate the common spatial misalignment and blurring in real imaging systems. Random displacement in different directions and amplitudes can enhance the ability of the network to simulate different fuzzy degradation. Subsequently, the feature maps with the same displacement direction and amplitude are connected to form multiple sets of feature representations with different degradation factors. Meanwhile, group convolution operations are performed on features of different groups. The above operations can be expressed as follows:

$$F_{Spa}^{D} = SC\left(GConv_{3\times3}\left(Cat_D\left(SO_{\phi_i}^{D}\left(DSConv\left(\left(X_f^{(t)}\right)\right)\right)\right)\right)\right) \tag{22}$$

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TABLE I
AVERAGE QUANTITATIVE PANSHARPENING RESULTS ON THE PAVIA CENTER, BOTSWANA, AND CHIKUSEI DATASETS

| Dataset | Methods | PSNR (↑) | SAM (↓) | SSIM (↑) | RMSE×$10^{-2}$ (↓) | ERGAS (↓) |
|---|---|---|---|---|---|---|
| Pavia center | CNMF [44] | 32.0244 | 7.2364 | 0.8913 | 2.7090 | 4.8195 |
| | HyperPNN [45] | 37.0370 | 5.2954 | 0.9520 | 1.5656 | 2.9419 |
| | DHP-DARN [48] | 37.1592 | 5.2685 | 0.9545 | 1.5299 | 2.8876 |
| | DIP-HyperKite [46] | 36.9586 | 5.3630 | 0.9527 | 1.5661 | 2.9363 |
| | Hyper-DSNet [27] | 37.1725 | 5.1541 | 0.9539 | 1.5245 | 2.8929 |
| | GuideNet [47] | 37.3485 | 5.2465 | 0.9529 | 1.5211 | 2.8542 |
| | HyperRefiner [28] | 37.7193 | 5.0461 | 0.9588 | 1.4265 | 2.7748 |
| | Tree-SNet [30] | 38.1569 | 4.8628 | 0.9594 | 1.3815 | 2.6432 |
| | DCTransformer [31] | 38.4304 | 4.8032 | 0.9626 | 1.3239 | 2.5836 |
| | FPFNet [58] | 37.2121 | 5.3962 | 0.9520 | 1.5297 | 2.8992 |
| | DFCFN [32] | 38.5797 | 4.7225 | 0.9621 | 1.3112 | 2.5485 |
| | FMPM-DNet [29] | 38.8193 | 4.6083 | 0.9633 | 1.2829 | 2.4837 |
| | Ours | **39.4058** | **4.4289** | **0.9662** | **1.2041** | **2.3646** |
| Botswana | CNMF [44] | 33.6006 | 2.3018 | 0.9193 | 4.5080 | 3.8151 |
| | HyperPNN [45] | 42.5861 | 1.9909 | 0.9522 | 1.3955 | 1.8251 |
| | DHP-DARN [48] | 41.1629 | 1.9877 | 0.94533 | 1.3589 | 2.5892 |
| | DIP-HyperKite [46] | 42.4773 | 1.7728 | 0.9576 | 1.2897 | 1.8724 |
| | Hyper-DSNet [27] | 43.7004 | 1.8418 | 0.9554 | 1.2998 | 1.4724 |
| | GuideNet [47] | 45.1494 | 1.5985 | 0.9673 | 1.0665 | 1.3201 |
| | HyperRefiner [28] | 44.2455 | 1.6751 | 0.9663 | 1.1655 | 1.3997 |
| | Tree-SNet [30] | 45.2278 | 1.5787 | 0.9671 | 1.0738 | 1.3039 |
| | DCTransformer [31] | 44.9387 | 1.5483 | 0.9671 | 1.0546 | 1.3320 |
| | FPFNet [58] | 44.1665 | 1.8960 | 0.9621 | 1.2396 | 1.4639 |
| | DFCFN [32] | 44.9590 | 1.5767 | 0.9687 | 1.0830 | 1.3166 |
| | FMPM-DNet [29] | 45.6029 | 1.5171 | 0.9692 | 1.0343 | 1.2521 |
| | Ours | **45.8482** | **1.4505** | **0.9702** | **0.9909** | **1.1986** |
| Chikusei | CNMF [44] | 35.6701 | 3.7933 | 0.8970 | 1.9730 | 6.6018 |
| | HyperPNN [45] | 40.6471 | 2.5668 | 0.9644 | 1.0318 | 4.5882 |
| | DHP-DARN [48] | 41.0419 | 2.4217 | 0.9686 | 0.9837 | 4.2614 |
| | DIP-HyperKite [46] | 41.6503 | 2.3207 | 0.9702 | 0.9246 | 4.0896 |
| | Hyper-DSNet [27] | 41.6990 | 2.3315 | 0.9705 | 0.9197 | 4.0150 |
| | GuideNet [47] | 42.7188 | 2.2169 | 0.9774 | 0.7961 | 3.6830 |
| | HyperRefiner [28] | 42.9958 | 2.1112 | 0.9777 | 0.8046 | 3.5126 |
| | Tree-SNet [30] | 43.2155 | 2.0608 | 0.9795 | 0.7487 | 3.4806 |
| | DCTransformer [31] | 43.2167 | 2.0527 | 0.9802 | 0.7450 | 3.4965 |
| | FPFNet [58] | 42.3698 | 2.3260 | 0.9759 | 0.8243 | 3.8711 |
| | DFCFN [32] | 43.4760 | 1.9978 | 0.9792 | 0.7680 | 3.3169 |
| | FMPM-DNet [29] | 44.0595 | 1.9074 | 0.9826 | 0.6809 | 3.1860 |
| | Ours | **44.3461** | **1.8458** | **0.9836** | **0.6609** | **3.1026** |

where $SO^D_{\phi_i}(\cdot)$ denotes the shift operation with amplitude $\phi_i$ in direction $D$, and $Cat_D(\cdot)$ denotes the operation of connecting feature maps with the same displacement direction and amplitude. $GConv_{3\times3}(\cdot)$ denotes the group convolution with a kernel size of $3 \times 3$, and $SC(\cdot)$ denotes the channel shuffle operation.

Finally, to enhance spectral adaptability during spatial degradation, we introduce a channel attention mechanism that captures global spectral dependencies. A depthwise separable convolution is first applied to integrate the features extracted by the preceding operations and restore the original channel dimensionality. Subsequently, average pooling and max pooling are performed across the spatial dimensions to generate two global descriptors encoding the overall channels. These descriptors are then processed by a shared fully connected network to produce two-channel attention vectors $W^D_{Avg}$ and $W^D_{Max}$, which are used to adaptively weight the integrated features $F^D_{Spa}$ obtained by the previous depthwise separable convolution. Two weighted features are added, and the feature maps are downsampled by a scaling factor of 4 to obtain the output $X^D_{Spa}$ of SpaD

$$\begin{cases} W^D_{Max} = S\left(MLP\left(GMP\left(DSConv\left(F^D_{Spa}\right)\right)\right)\right) \\ W^D_{Avg} = S\left(MLP\left(GAP\left(DSConv\left(F^D_{Spa}\right)\right)\right)\right) \end{cases} \tag{23}$$

$$X^D_{Spa} = \left(W^D_{Avg} \odot F^D_{Spa} + W^D_{Max} \odot F^D_{Spa}\right) \downarrow_4 \tag{24}$$

where $GMP(\cdot)$ represents the global max pooling, and $MLP(\cdot)$ represents a fully connected network.

In SpaR, the difference maps between the output of SpaD and the input LRHS image are first upsampled and then processed by the operations, which are identical to those in SpaD, as shown in (22)–(24). The processed features, i.e., the output of SpaR, are mapped to high-resolution space.

*4) SpeD/SpeR Blocks:* In the process of solving HS pansharpening problems, another key operation is to model the spectral degradation and reconstruction process, which plays a decisive role in preserving the spatial–spectral fidelity of the fused HRHS image. A spatial–spectral aware SpeD and SpeR are constructed to achieve spectral degradation and reconstruction by effectively capturing the relationships between spectral

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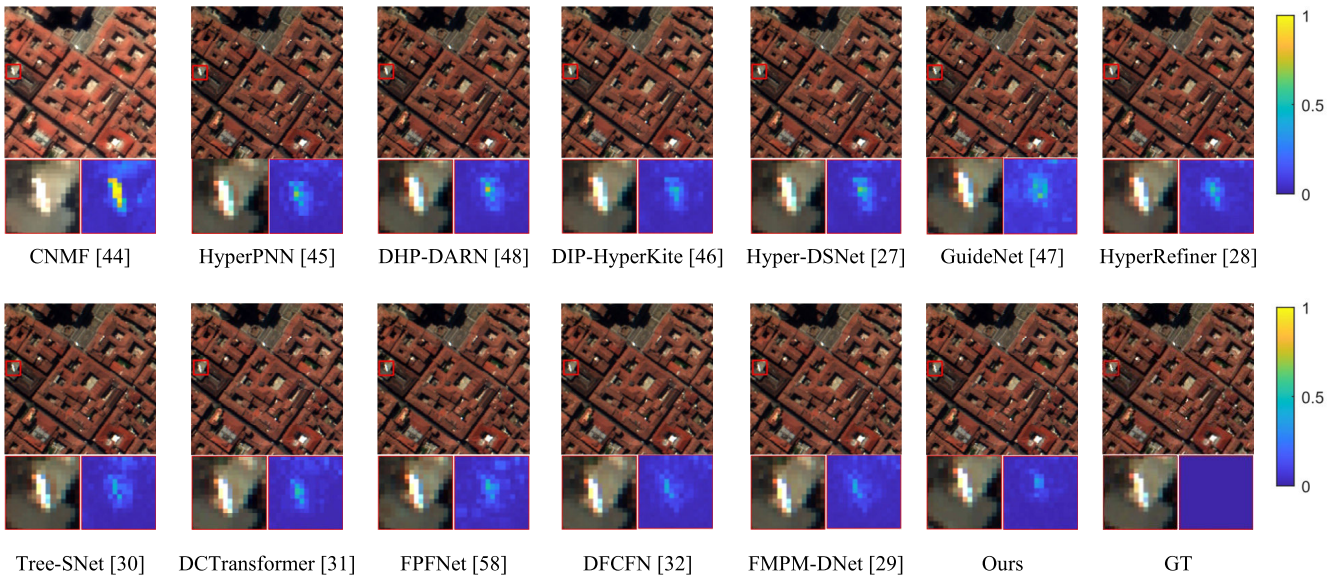



Fig. 3. Fusion results on the Pavia Center dataset.

channels in multiple frequency components. As shown in Fig. 1(d), similar to spatial degradation and reconstruction, SpeD and SpeR also use the same structure and different inputs to achieve spectral feature degradation and reconstruction. Next, we take SpeD as an example to introduce its specific structure.

SpeD takes the fused HRHS image $X_f^{(t)}$ from CMFU as input to learn the spectral degradation process between the multiband image and single-band image. To better represent the complex spectral degradation process, we introduce a spatial–spectral aware attention mechanism to adaptively learn the contribution of each frequency component. Specifically, a discrete wavelet transform (DWT) is adopted to decompose the HS image $X_f^{(t)}$ into four frequency components $[ll, lh, hl, hh]$. Four frequency components are concatenated and sent to the spatial-aware branch and spectral-aware branch, respectively. In the spatial-aware branch, a $3 \times 3$ convolution and a Sigmoid function are used to integrate spatial information and obtain spatial attention weights $W_{Spa}^D$. In the spectral-aware branch, channel attention weights $W_{Spe}^D$ are obtained by capturing the feature correlations of different frequency components in the channel dimension. Spatial and channel attention weights are used to weight the four components in the wavelet domain, and then the inverse DWT (IDWT) is performed to transform the image features back into the spatial domain, thereby obtaining intermediate degraded features $F_{Spe}^D$. The above process is formulated as follows:

$$[ll, lh, hl, hh] = DWT\left(X_f^{(t)}\right), F_c = Cat\left(ll, lh, hl, hh\right) \tag{25}$$

$$\begin{cases} W_{Spa}^D = S\left(Conv_{3\times3}\left(F_c\right)\right) \\ W_{Spe}^D = S\left(Conv_{1\times1}\left(ReLu\left(Conv_{1\times1}\left(GAP\left(F_c\right)\right)\right)\right)\right) \end{cases} \tag{26}$$

$$F_{Spe}^D = IDWT\left(W_{Spe}^D \odot W_{Spa}^D \odot F_c\right) \tag{27}$$

where $DWT(\cdot)$ and $IDWT(\cdot)$ represent the operations of DWT and IDWT, respectively, and $F_c$ represents the cascading features of four frequency components $[ll, lh, hl, hh]$. Subsequently, a dynamic gating fusion mechanism is designed to integrate the input spectral features with the degraded features to achieve adaptive modulation of the degraded features. Finally, a $1 \times 1$ convolution is used to integrate features and obtain a single-band degraded image $X_{Spe}^D$

$$X_{Spe}^D = Conv_{1\times1}\left(\varepsilon F_{Spe}^D + (1-\varepsilon)\left(Conv_{3\times3}\left(X_f^{(t)}\right)\right)\right). \tag{28}$$

SpeR aims to reconstruct a multiband image from a single-band image by accurately restoring spectral and spatial information. The operations in SpeR are identical to those in SpeD, as shown in (25)–(28). Note that the input of SpeR is a single-band image, which is different from the input of SpeD.

### C. Loss Function

For network training, we define a joint loss function consisting of a reconstruction loss $L_{rec}$ and a spectral similarity loss $L_{Spe}$, as shown below

$$L = L_{rec} + \rho L_{Spe} \tag{29}$$

where $\rho$ is a trade-off parameter that balances the contribution of reconstruction loss and spectral similarity loss.

$L_1$ loss is used to define the reconstruction loss $L_{rec}$, which has proven effective for HS pansharpening tasks. The definition of $L_{rec}$ is as follows:

$$L_{rec} = \frac{1}{CHW}\left\|I_{GT} - \hat{I}_{HRHS}\right\|_1 \tag{30}$$

where $I_{GT}$ denotes the ground-truth (GT) image, and $\hat{I}_{HRHS}$ denotes the generated HRHS image, i.e., the reconstructed image $X$ obtained in (12).

To ensure the spectral fidelity of fused HRHS images, a spectral similarity loss is defined to measure spectral similarity by calculating the cosine value between the spectral vectors at each position in the GT and generated HRHS images. The closer the two spectral vectors are at position $k$, the closer the

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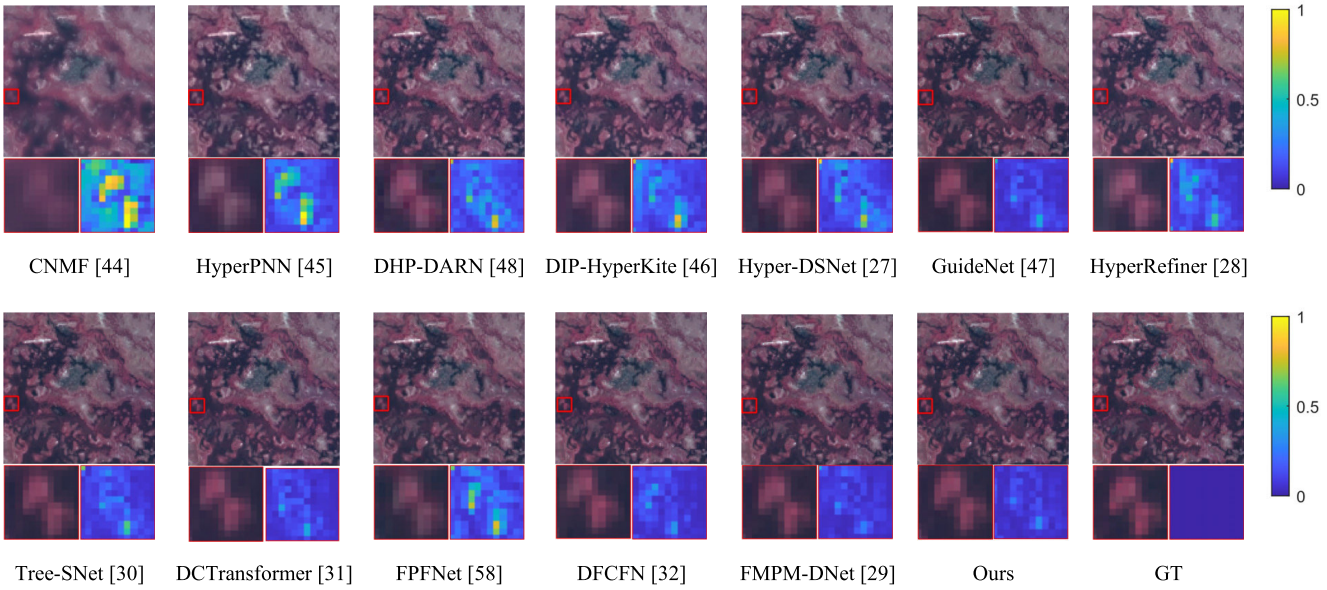



Fig. 4. Fusion results on the Botswana dataset.

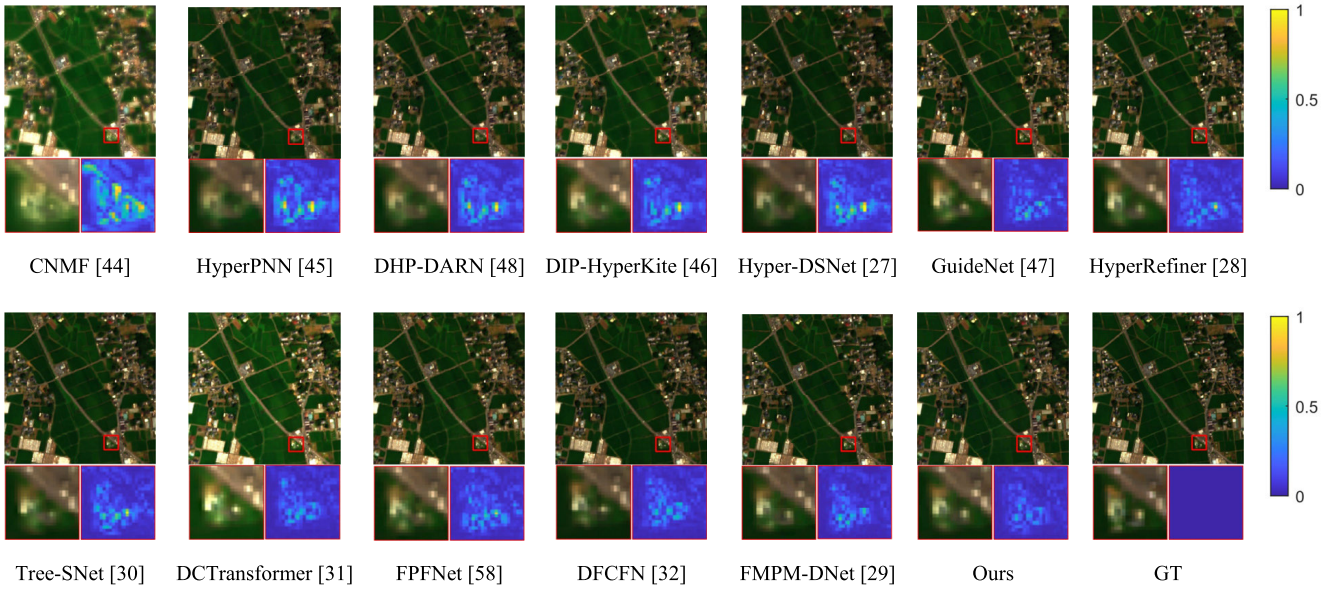



Fig. 5. Fusion results on Chikusei dataset.

corresponding cosine value $\cos\theta_k$ is to 1. To ensure network convergence, the spectral similarity loss $L_{Spe}$ is defined as follows:

$$L_{Spe} = 1 - \frac{1}{HW}\sum_{k=1}^{HW}\cos\theta_k \tag{31}$$

$$\cos\theta_k = \frac{I_{GT}^k \cdot \hat{I}_{HRHS}^k}{\left\|I_{GT}^k\right\|_2 \cdot \left\|\hat{I}_{HRHS}^k\right\|_2}. \tag{32}$$

## IV. Experiments

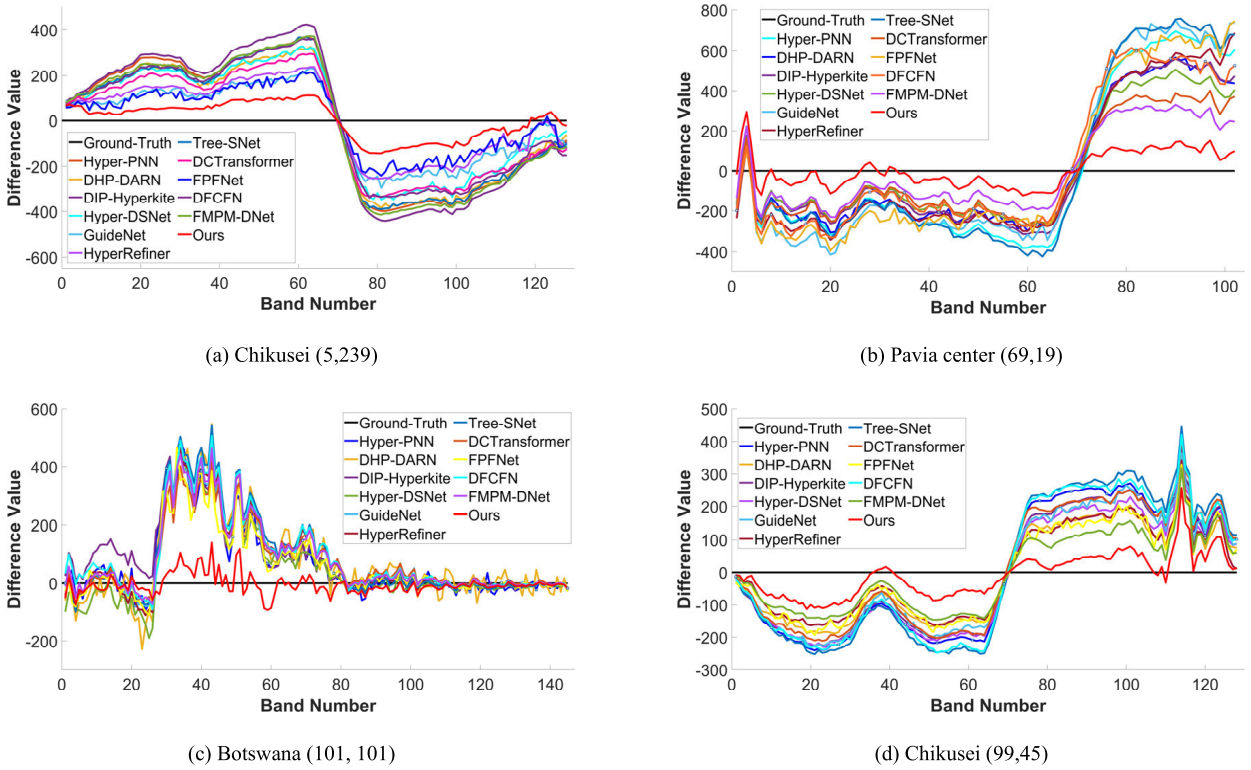
### A. Datasets and Experimental Setup

To evaluate the comprehensive performance of the proposed 3SP-DUNet, we conducted extensive experiments on three widely used simulated benchmark datasets: Pavia Center dataset [54], Botswana dataset [55], and Chikusei dataset [56], as well as one real-world dataset: FR1 dataset [27], and compared it with a series of SOTA methods.

*1) Pavia Center Dataset:* The original data from the Pavia Center scene was acquired by the reflective optics system imaging spectrometer (ROSIS). 13 noise spectral bands and a central rectangular region of $1096 \times 381$ without information were removed from the original data to obtain a "two-part" image with the size of $1096 \times 715 \times 102$. The image was segmented into 24 nonoverlapping cubes with dimensions of $160 \times 160 \times 102$, which serve as reference (GT) images in the dataset. Following Wald's protocol [57] and the experimental settings outlined in [46], corresponding PAN and LRHS images with sizes of $160 \times 160$ and $40 \times 40 \times 102$ were generated, respectively. Among the 24 sets of images, 17 sets were used for training, and the remaining seven sets were used for testing.

*2) Botswana Dataset:* The Botswana data was acquired by the Hyperion sensor aboard the National Aeronautics

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(a) Chikusei (5,239)

(b) Pavia center (69,19)

(c) Botswana (101, 101)

(d) Chikusei (99,45)

Fig. 6. Spectral differences at four locations in Figs. 3–5 between the GT and pansharpening results. (a) Chikusei (5239). (b) Pavia center (6919). (c) Botswana (101 101). (d) Chikusei (9945).

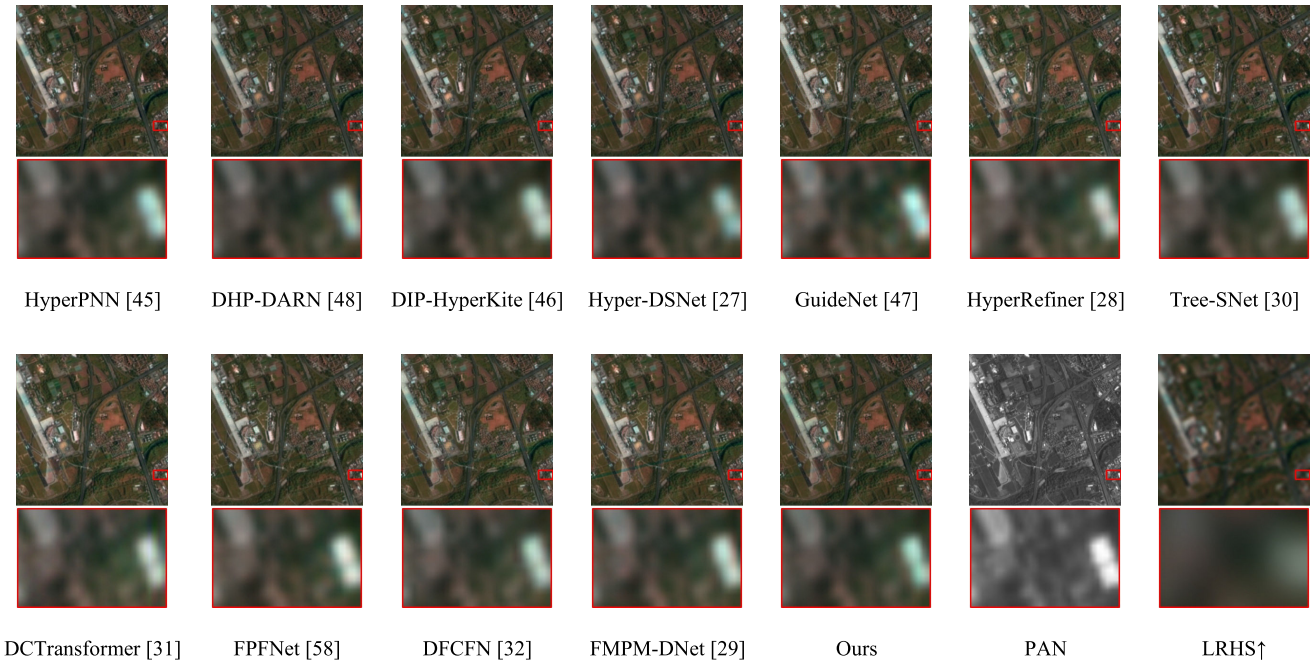



Fig. 7. Fusion results on the real FR1 dataset.

and Space Administration Earth Observing-1 (EO-1) satellite over a site in Botswana between 2001 and 2004. This data contains 242 spectral bands spanning from 400 to 2500 nm, with a spatial size of 1496 × 256 pixels. The uncalibrated and noisy spectral bands were removed from the original data to obtain an HSI with 145 spectral bands. The HSI is further cut into 24 nonoverlapping cube blocks with a size of $120 \times 120 \times 145$ to form reference (GT) images in the Botswana dataset. Similarly, following Wald's protocol [57] and the experimental settings outlined in [46], corresponding PAN and LRHS images with sizes of $120 \times 120$ and $30 \times 30 \times 145$ were obtained, respectively. 18 sets of images were randomly selected for training, while the remaining sets were used for testing.

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TABLE II
AVERAGE QUANTITATIVE ON THE REAL FR1 DATASET

| Methods | $D_\lambda$ (↓) | $D_s$ (↓) | QNR (↑) |
|---|---|---|---|
| CNMF [44] | 0.0253 | 0.0283 | 0.9470 |
| HyperPNN [45] | 0.0331 | 0.0309 | 0.9369 |
| DHP-DARN [48] | 0.0249 | 0.0216 | 0.9539 |
| DIP-HyperKite [46] | 0.0272 | 0.0260 | 0.9474 |
| Hyper-DSNet [27] | 0.0259 | 0.0238 | 0.9508 |
| GuideNet [47] | 0.0334 | 0.0282 | 0.9393 |
| HyperRefiner [28] | 0.0257 | 0.0242 | 0.9505 |
| Tree-SNet [30] | 0.0335 | 0.0243 | 0.9335 |
| DCTransformer [31] | 0.0255 | 0.0237 | 0.9514 |
| FPFNet [58] | 0.0394 | 0.0327 | 0.9293 |
| DFCFN [32] | 0.0227 | 0.0201 | 0.9575 |
| FMPM-DNet [29] | 0.0226 | 0.0205 | 0.9574 |
| Ours | **0.0198** | **0.0165** | **0.9641** |

TABLE III
ABLATION STUDY RESULT OF SPED/SPER AND SPAD/SPAR ON THE PAVIA CENTER AND BOTSWANA DATASETS

| SpaD/SpaR | SpeD/SpeR | PSNR (↑) | SAM (↓) | SSIM (↑) | RMSE×$10^{-2}$ (↓) | ERGAS (↓) |
|---|---|---|---|---|---|---|
| × | × | 38.7782 | 4.6756 | 0.9630 | 1.2898 | 2.4891 |
| | | 44.6846 | 1.6354 | 0.9646 | 1.0988 | 1.3625 |
| ✓ | × | 38.7957 | 4.7043 | 0.9635 | 1.2885 | 2.4816 |
| | | 45.2249 | 1.5526 | 0.9674 | 1.0518 | 1.2899 |
| × | ✓ | 39.1394 | 4.5030 | 0.9651 | 1.2398 | 2.4173 |
| | | 45.0605 | 1.5664 | 0.9670 | 1.0539 | 1.3195 |
| ✓ | ✓ | **39.4058** | **4.4289** | **0.9662** | **1.2041** | **2.3646** |
| | | **45.8482** | **1.4505** | **0.9702** | **0.9909** | **1.1986** |

Note: The first and second rows correspond to the Pavia center and Botswana datasets, respectively.

TABLE IV
ABLATION STUDY RESULT OF SPECTRAL/SPATIAL SPARSE PRIOR ON THE PAVIA CENTER AND BOTSWANA DATASETS

| Methods | PSNR(↑) | SAM(↓) | SSIM(↑) | RMSE×$10^{-2}$ (↓) | ERGAS(↓) |
|---|---|---|---|---|---|
| w/o Spectral/Spatial Sparse Prior | 39.1214 | 4.5016 | 0.9650 | 1.2417 | 2.4192 |
| | 44.4624 | 1.6831 | 0.9643 | 1.1177 | 1.4231 |
| w/ Spectral/Spatial Sparse Prior | **39.4058** | **4.4289** | **0.9662** | **1.2041** | **2.3646** |
| | **45.8482** | **1.4505** | **0.9702** | **0.9909** | **1.1986** |

Note: The first and second rows correspond to the Pavia center and Botswana datasets, respectively.

TABLE V
ABLATION STUDY RESULT OF CMFU STRUCTURE ON THE PAVIA CENTER AND BOTSWANA DATASETS

| Methods | PSNR (↑) | SAM (↓) | SSIM (↑) | RMSE×$10^{-2}$ (↓) | ERGAS (↓) |
|---|---|---|---|---|---|
| w/o CMFU | 39.1000 | 4.5099 | 0.9648 | 1.2459 | 2.4220 |
| | 45.2592 | 1.5436 | 0.9677 | 1.0426 | 1.2828 |
| w/ CMFU | **39.4058** | **4.4289** | **0.9662** | **1.2041** | **2.3646** |
| | **45.8482** | **1.4505** | **0.9702** | **0.9909** | **1.1986** |

Note: The first and second rows correspond to the Pavia center and Botswana datasets, respectively.

*3) Chikusei Dataset:* The Chikusei data were obtained by the Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban areas in Chikusei, Japan. This data contains 128 spectral bands spanning from 363 to 1018 nm, with a spatial size of $2517 \times 2335$ pixels. We removed invalid edge pixels from the data and cut it into 72 nonoverlapping cube blocks of size $256 \times 256 \times 128$. Similarly, following Wald's protocol [57] and the experimental settings outlined in [46], corresponding PAN and LRHS images with sizes of $256 \times 256$ and $64 \times 64 \times 128$ were obtained, respectively. Among the 72 sets of images, 54 sets of images were used for training, and the remaining 18 sets were used for testing.

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TABLE VI
ABLATION STUDY RESULT OF SPECTRAL SIMILARITY LOSS ON THE PAVIA CENTER AND BOTSWANA DATASETS

| Methods | PSNR (↑) | SAM (↓) | SSIM (↑) | RMSE×$10^{-2}$ (↓) | ERGAS (↓) |
|---|---|---|---|---|---|
| w/o Spectral Loss | 38.8798 | 4.6238 | 0.9637 | 1.2729 | 2.4698 |
| | 45.1222 | 1.5437 | 0.9676 | 1.0429 | 1.3217 |
| w/ Spectral Loss ($\rho = 1$) | 38.1647 | 4.5484 | 0.9580 | 1.4423 | 2.6990 |
| | 44.1622 | 1.6711 | 0.9625 | 1.1578 | 1.4394 |
| w/ Spectral Loss ($\rho = 0.1$) | **39.4058** | **4.4289** | **0.9662** | **1.2041** | **2.3646** |
| | **45.8482** | **1.4505** | **0.9702** | **0.9909** | **1.1986** |
| w/ Spectral Loss ($\rho = 0.01$) | 39.1760 | 4.5000 | 0.9654 | 1.2323 | 2.4088 |
| | 45.0095 | 1.5857 | 0.9663 | 1.0648 | 1.3186 |

Note: The first and second rows correspond to the Pavia center and Botswana datasets, respectively.

*4) FR1 Dataset:* The FR1 data, initially released for the PRISMA contest, provides precisely co-registered image pairs consisting of a PAN image with a size of 2400 × 2400 and an LRHS image with a size of 400 × 400 × 69. The inherent spatial resolution ratio between the PAN and LRHS images is 6:1. To conduct experiments at the common ratio of 4:1, we first downsampled the PAN image from the original resolution of 2400×2400 to a resolution of 1600×1600. Then, following Wald's protocol [57] and the experimental settings outlined in [46], the LRHS and PAN images were downsampled by a factor of four times, and the original LRHS image was used as the GT image. The cutting operation was performed on three types of images to obtain 25 pairs of images, each consisting of a GT image (80 × 80 × 69), a corresponding LRHS image (20 × 20 × 69), and a PAN image (80 × 80 pixels). 18 pairs of images were randomly selected for training, while the remaining images were used for testing. In addition, to further validate the generalization of the model, we performed cutting operations on full-resolution image pairs (i.e., LRHS with a size of 400 × 400 × 69 and PAN with a size of 1600 × 1600) to obtain six sets of source image pairs consisting of LRHS images with a size of 80 × 80 × 69 and PAN images with a size of 320 × 320. These source image pairs are used as real data to validate the performance of the model.

For fair comparison, all DL-based methods were uniformly reimplemented using Python 3.9 with PyTorch 1.13, and trained on an NVIDIA L20 GPU under the Ubuntu 20.04 operating system. All models were trained for 5000 epochs using the Adam optimizer with a batch size of 1. The initial learning rate was set to 0.0001 and decayed by a factor of 0.8 every 2000 epochs. The spatial/spectral sparse prior in our model was optimized 50 times through ISTA using a sparsity regularization parameter of 0.1.

## B. Comparison Methods and Metrics

To verify the effectiveness of 3SP-DUNet, we compared the proposed model with several SOTA methods, including one traditional method, CNMF [44], and 11 DL-based methods, such as HyperPNN [45], DHP-DARN [48], DIP-HyperKite [46], Hyper-DSNet [27], GuideNet [47], HyperRefiner [28], Tree-SNet [30], DCTransformer [31], FPFNet [58], DFCFN [32], and FMPM-DNet [29]. Six quantitative metrics were used to evaluate the performance of comparative methods, such as spectral angle mapping (SAM) [59], structural similarity (SSIM) [60], root mean square error (RMSE) [59], erreur relative globale adimensionnelle de synthese (ERGAS) [59], and peak signal-to-noise ratio (PSNR) [60]. Furthermore, three objective metrics were used in the real experiment: quality with no reference (QNR), $D_\lambda$, and $D_s$.

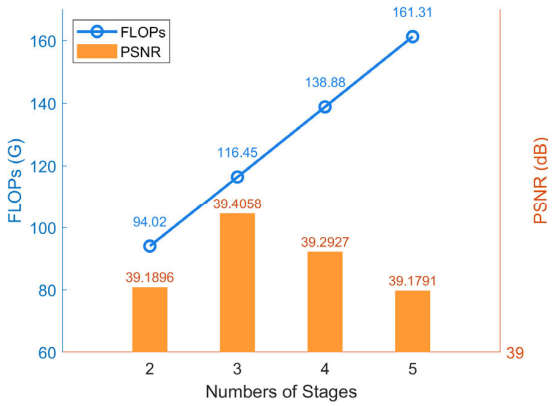



Fig. 8. Ablation study of the number of stages on the Pavia Center dataset.

## C. Experimental Results

*1) Experimental Results on Simulated Datasets:* Table I shows the objective evaluation results obtained by all comparison methods on the three datasets. The best results are highlighted in bold, while the second-best results are marked with underlines. As shown in the table, the proposed method achieves the highest PSNR values among all compared approaches. A higher PSNR indicates that the fusion result is closer to the GT image, with less geometric distortion. In addition, our method attains the lowest SAM values, where a smaller SAM reflects a reduction in spectral distortion. Overall, these results demonstrate that our method achieves superior spatial and spectral fidelity.

Figs. 3–5 show the subjective comparison of fusion results obtained by different methods on the Pavia Center, Botswana, and Chikusei datasets, respectively. To generate the RGB images, the 10th, 11th, and 12th spectral bands are used as the blue channel; the 30th, 35th, and 20th bands as the green channel; and the 60th, 61st, and 29th bands as the red channel for the Pavia Center, Botswana, and Chikusei datasets, respectively. To more clearly illustrate the differences among the fusion results, we computed and visualized the mean absolute error (MAE) maps between the fused images and the corresponding GT images. In each figure, the bottom-

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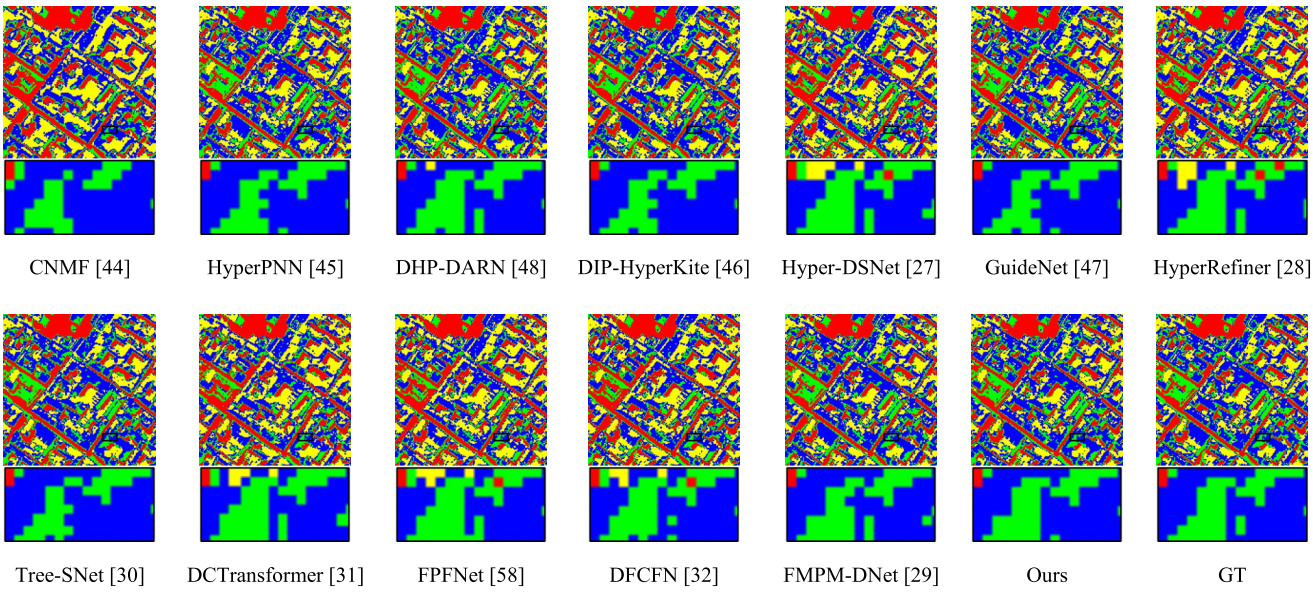



Fig. 9. Classification results on the Pavia Center dataset.

left corner displays an enlarged local region of the fusion result, while the bottom-right corner shows the corresponding enlarged MAE map. As observed in the figures, the MAE maps produced by our method exhibit deeper blue tones and fewer residuals, indicating that our fusion results are closer to the GT images.

To conduct a more in-depth comparison of the spectral retention performance exhibited by DL-based approaches, Fig. 6 presents the spectral difference curves of four randomly selected pixels in three images from these datasets, which are used to assess the spectral preservation capability of DL-based methods. Our method has the smallest spectral difference values compared to other methods, showing excellent spectral preservation capability.

*2) Experimental Results on Real Dataset:* We conducted experiments on the real dataset FR1. As illustrated in Fig. 7, the qualitative results indicate that our method produces the fused image with sharper edges and richer texture details. In contrast, other methods tend to suffer from noticeable blurring in fine details. As reported in Table II, our approach achieves the highest scores across all no-reference evaluation metrics, confirming its strong generalization on the FR1 dataset.

### D. Ablation Study

*1) Effectiveness of SpeD/SpeR and SpaD/SpaR:* To verify the effectiveness of SpeD/SpeR and SpaD/SpaR, we replace them with $3 \times 3$ convolution operations in the network. Table III shows the results of different variants produced by replacing SpeD/SpeR or SpaD/SpaR with convolution operations. The results indicate that our model is significantly superior to variants without these modules.

*2) Effectiveness of Spectral/Spatial Sparse Prior:* To validate the efficacy of spectral and spatial sparse prior blocks, we replace them with residual blocks consisting of a $3 \times 3$ convolution, a ReLU function, and a $3 \times 3$ convolution. The quantitative results shown in Table IV indicate that the network using spectral and spatial sparse prior blocks outperforms that using residual blocks, which also verifies the good ability of these two blocks to capture spectral and spatial information.

*3) Effectiveness of CMFU:* Table V shows the ablation results of networks with and without CMFU. Quantitative assessments demonstrate that the CMFU significantly mitigates the degradation of spectral and spatial fidelity. Specifically, by preserving and propagating reconstructed features across successive iterations, CMFU effectively compensates for the lost information that typically arises as the unfolding architecture deepens.

*4) Effectiveness of Spectral Similarity Loss:* Table VI shows the ablation results with and without spectral similarity loss, as well as the results of setting different hyperparameters for spectral similarity loss. The quantitative results indicate that the 3SP-DUNet trained with spectral similarity loss achieves superior fusion performance when $\rho$ is set to 0.1, which also verifies the key role of spectral similarity loss in spectral fidelity. Conversely, assigning a high weight to the spectral similarity loss may suppress high-frequency components, thereby compromising the reconstruction of spatial details and leading to blurred textures. For example, when $\rho = 1$, all indicators show a significant decrease compared to indicators without spectral similarity loss.

*5) Number of Stages:* To determine the optimal number of stages in 3SP-DUNet, Fig. 8 shows the performance and computational efficiency of networks with different numbers of stages. The results show that the three unfolding stages achieve the best fusion performance.

*6) Model Complexity:* We compare the parameter count and computational complexity across seven DL-based models. Table VII shows that our method achieves SOTA performance with the least number of parameters and a compromise in computational complexity. Furthermore, we evaluated the inference time of the compared models. It is important to note that FLOPs are not always directly proportional to inference time, especially given the parallel processing nature of GPUs. Although our model does not have the lowest FLOPs, it offers

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TABLE VII
COMPARISON OF PARAMETERS AND FLOPS OF DL-BASED METHODS ON THE CHIKUSEI DATASET

| Methods | GuideNet [47] | HyperRefiner [28] | Tree-SNet [30] | DCTransformer [31] | FPFNet [58] | DFCFN [32] | FMPM-DNet [29] | Ours |
|---|---|---|---|---|---|---|---|---|
| #Params (M) | 5.13 | 19.32 | 9.12 | 18.50 | 22.34 | 10.95 | 5.84 | 3.09 |
| FLOPs (G) | 150.17 | 85.00 | 205.00 | 3220.19 | 174.92 | 83.43 | 100.16 | 300.63 |
| Time (ms) | 9.82 | 2371.10 | 899.49 | 1495.03 | 22.71 | 53.93 | 57.45 | 131.04 |
| PSNR (↑) | 42.7188 | 42.9958 | 43.2155 | 43.4759 | 42.3698 | 43.4759 | 44.0595 | **44.3092** |

TABLE VIII
OBJECTIVE COMPARISON OF CLASSIFICATION RESULTS ON THE PAVIA CENTER DATASET

| Metrics | Hyper-DSNet [27] | GuideNet [47] | HyperRefiner [28] | Tree-SNet [30] | DCTransformer [31] | FPFNet [58] | DFCFN [32] | FMPM-DNet [29] | Ours |
|---|---|---|---|---|---|---|---|---|---|
| OA | 0.9435 | 0.9501 | 0.9465 | 0.9519 | 0.9480 | 0.9459 | 0.9526 | 0.9564 | **0.9577** |
| Kappa | 0.9225 | 0.9315 | 0.9265 | 0.9340 | 0.8287 | 0.9258 | 0.9350 | 0.9402 | **0.9420** |

a competitive balance between inference speed, parameter size, and accuracy.

## V. APPLICATION

To validate the effectiveness of our proposed 3SP-DUNet in downstream tasks, we performed a downstream land-cover classification experiment on the fused results from the Pavia Center dataset. The K-Means clustering algorithm (four classes, five iterations) was employed on the outputs of all compared pansharpening methods. Table VIII shows that the result generated by our 3SP-DUNet achieves the highest classification accuracy. Fig. 9 shows the visualization results for different fusion results, and we can observe that the classification result of our result is closest to that on GT. This experiment demonstrates that our method yields results that are closest to GT, indicating better spectral and spatial fidelity.

## VI. CONCLUSION

This article proposes an interpretable DUN called 3SP-DUNet for the HS pansharpening task. The network is designed by unfolding a model-based optimization process that explicitly simulates spectral and spatial degradation and reconstruction. In this network, SpeD/SpeR and spatial SpaD/SpaR modules are designed to model the degradation and reconstruction processes of images. To further enhance the fidelity of fused features, two sparse prior blocks based on sparse representation learning are integrated into the network. Furthermore, to reduce information loss during the iterative process, CMFU is constructed to fuse reconstructed images from adjacent stages. The entire network is optimized using a joint loss function that consists of a reconstruction term and a spectral similarity constraint. Extensive experiments on the Pavia Center, Botswana, Chikusei, and FR1 datasets demonstrate that the proposed 3SP-DUNet outperforms SOTA methods in both quantitative metrics and visual effects. Although our current framework offers theoretical transparency, it relies on a single-step upsampling strategy. This structural design may limit the model's capability to fully capture the intricate spectral and spatial hierarchies inherent in the data. Consequently, future work will focus on developing a DUN grounded in multiscale interpretable theories, aimed at improving the fidelity of reconstructed features by exploiting cross-scale information.

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**Yong Yang** (Senior Member, IEEE) received the Ph.D. degree from Xi'an Jiaotong University, Xi'an, China, in 2005.

From 2009 to 2010, he was a Post-Doctoral Research Fellow at Chonbuk National University, Jeonju, South Korea. He is currently a Distinguished Professor with the School of Computer Science and Technology, Tiangong University, Tianjin, China. His research interests include image fusion, image super-resolution reconstruction, medical image processing and analysis, and deep learning.

Dr. Yang is an Associate Editor of IEEE ACCESS and an Editor of *KSII Transactions on Internet and Information Systems*.



**Xuan Liu** received the B.E. degree in automation from Tiangong University, Tianjin, China, in 2024, where he is currently pursuing the M.S. degree in control science and engineering.

His research interests include computer vision, machine learning, and remote sensing image fusion.



**Shuying Huang** (Member, IEEE) received the Ph.D. degree in computer application technology from the Ocean University of China, Qingdao, China, in 2013.

She is currently a Professor with the School of Software, Tiangong University, Tianjin, China. Her research interests include image and signal processing and pattern recognition.



**Xiaozheng Wang** received the B.S. degree in mechanical design, manufacturing, and automation from North Minzu University, Ningxia, China, in 2018. He is currently pursuing the Ph.D. degree in control science and engineering with Tiangong University, Tianjin, China.

His research interests include computer vision, machine learning, and remote sensing image fusion.



**Jinlin Lu** received the B.S. degree in control engineering from Tiangong University, Tianjin, China, in 2025, where he is currently pursuing the Ph.D. degree in control science and engineering.

His research interests include computer vision, machine learning, image super-resolution, and remote sensing image fusion.

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